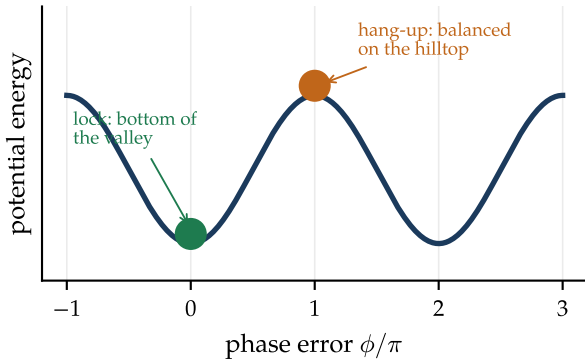


(a) no frequency offset: $U = -\cos \phi$



(b) frequency offset tilts the washboard

